Agentic AI Bootcamp

Research Workflows, Teaching & Applications

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thinkingwithagents.github.io

Where We Left Off & Today's Plan

Session 1

- Agentic AI vs. chat AI
- Context window & planning
- `CLAUDE.md`, `PLAN.md`, `SKILL.md`
- Permissions & safety
- Apps: voice file, website, lecture builder spec

Emily: Research

1. **Data pipeline** — curiosity to map in 20 min
2. **Code & paper audit** — AI vs. real referees

Multi-step workflows where you check each step.

Erkmen: Teaching

1. **Lecture Builder** — papers to slides
2. **Research Brainstorm** — stress-test live
3. **Web Scraper** — novel data in 10 min

Setup: Install Your Skill Toolkit

Anthropic skills (one-time)

1. Add the marketplace:

```
/plugin marketplace add anthropics/skills
```

2. Install the document-skills bundle:

```
/plugin install
  document-skills@anthropic-agent-skills
```

Gives you: `pdf`, `docx`, `pptx`, `xlsx`, `frontend-design`, `skill-creator`, `algorithmic-art`.

Third-party skills via `npx`

Full economist pack:

```
npx skills add thinkingwithagents/skills
```

One skill at a time:

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npx skills add thinkingwithagents/skills
  --skill research-brainstorm
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Includes: `research-brainstorm`, `find-data`, `academic-beamer-deck`.

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`/skill-creator` is **not** bundled with Claude Code by default — it ships inside `document-skills`. Install it before App 3.

Emily: Research Workflows

Data Pipelines & Code Auditing

From Chat to Workflows

Session 1 Recap

- Chat AI vs. agentic AI
- The context window
- Voice files & standing instructions
- Skills: reusable workflows

Today's theme:

Multi-step pipelines where AI does the mechanical parts and **you check each step**.

Describe $\rightarrow$ Propose $\rightarrow$ Choose $\rightarrow$ Execute $\rightarrow$ Review.

You stay in the driver's seat.

Application 1

Lead Contamination in Vermont

From Curiosity to Map in 20 Minutes

The Question

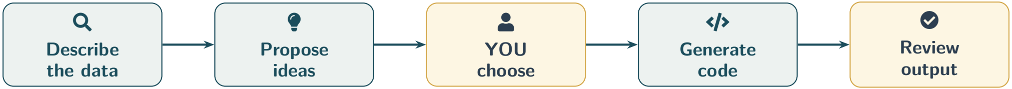
- Nick Saunders gave a talk on lead contamination
- I wanted to know: **what does lead look like in Vermont?**
- Downloaded TRI data from the EPA — Toxics Release Inventory, public data — filtered to Vermont and lead
- One CSV file. That's it.

This started from **genuine curiosity**, not a prepared exercise.

The Data

- 429 records
- 36 facilities
- 1987–2024 (38 years)
- 24 variables
- Unit: pounds of lead released

The Workflow: Describe, Propose, Choose, Code, Review



= AI does the mechanical work
= You provide judgment

Each step is **narrow enough to check**. You're not asking the AI to "do my research."

>_ LIVE DEMO

Step 1: Describe the Data

```
$ claude

Read vt_lead_tri.csv and give me a thorough description of
what's in it.  What variables are there, what's the coverage,
who are the top facilities, and anything else I should know
before analyzing it.
```

▶ Switching to terminal...

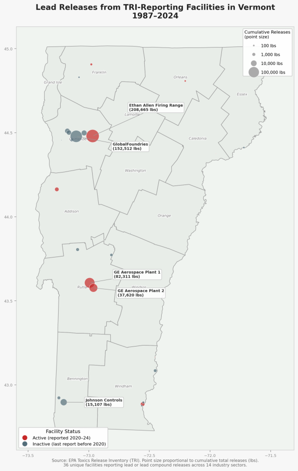
>_ LIVE DEMO

Step 2: Propose Analysis Ideas

```
Based on this data, propose 4 research or visualization ideas
I could pursue, ranging from a quick afternoon project to a
real research project.  For each one, tell me what it would
show, why it's interesting, what additional data I'd need,
and how complex it would be.
```

→ **Key line:** YOU choose what's interesting, not the AI. It generates options. You apply judgment.

The Map: Vermont Lead Releases by Facility



Each dot is a facility. Size = cumulative lead releases (lbs), 1987–2024. Source: EPA TRI.

Key Findings: Two Facilities Dominate

Facility	Lbs
Ethan Allen Firing Range	208,665
GlobalFoundries (Essex Jct)	152,512
GE Aerospace Plant 1 (Rutland)	82,311
GE Aerospace Plant 2	37,620
Johnson Controls (Bennington)	15,107

Top 2 facilities = **69%** of all lead releases.
Top 3 = **85%**.
Total across all facilities: 521,410 lbs.

The firing range is a National Guard site — lead from spent ammunition deposited directly into soil. A completely different pollution story than semiconductor manufacturing.

Time Trend

Before 2007: industrial off-site transfers.
After 2007: military on-site deposition.
The *nature* of pollution changed.

From Curiosity to Written Summary: ~20 Minutes

The Summary Memo

AI generated a full memo covering:

- Key findings and top facilities
- Time trends and compositional shifts
- Geographic concentration (2 counties = 95% of releases)
- A real limitations section

"TRI is self-reported, doesn't measure actual exposure, only covers facilities above reporting thresholds."

The full pipeline:

1. Describe the data
2. Propose analysis ideas
3. I chose the map
4. AI generated Python code
5. Reviewed the output
6. AI wrote the memo

Total time: ~20 minutes.
By hand: most of a day.

What AI Is Good At Here vs. Not

✓ AI Strengths

- **Data orientation** — reading a CSV, summarizing, catching quirks
- **Code generation** — the map worked on the first try
- **First-draft reporting** — memo structure, limitations section
- **Proposing angles** you might not think of

✗ Still Needs You

- **Vermont-specific knowledge** — Jericho is near residential areas
- **Policy context** — who uses this, what decisions it informs
- **Deciding what matters** — 4 ideas, can't tell you which matters for *your* work

The AI is a very fast, very thorough RA who **just arrived in Vermont last week**. It can do anything you ask, but it doesn't know what to ask.

Application 2

Code Audit & Paper Audit

Bangladesh Remote Education RCT

The Problem

- Paper on remote education in Bangladesh
- Evaluates TV instruction, adaptive edtech, data subsidies, and teacher support during COVID-era school closures
- **Keeps getting rejected** — 8 submissions, 11 referee reports
- Used AI two ways:
  - Clean up the code
  - Stress-test the paper

Two use cases:

1. Code audit
Restructure 3 years of accumulated code into a clean replication package

2. Paper audit
Adversarial review before resubmission — catch what referees would catch

Code Audit: Before

```
02_Do files/
01_master_5th June 2021.do
03_vargen_07sep.do
03_vargen.do
05_endline balance_eb.do
05_endline balance.do
09_attrition_8 July_e.do
09_attrition_8 July.do
...
Archive/ (18 files)
WB Submission Files/ (14 copies)
```

The problems:

- **51 do-files** across 4 subdirectories
- Master file from **June 2021** — 3 years stale
- Date-in-filename: `_07sep`, `_02July1130am`
- Author-in-filename: `_eb`, `_e`
- Hardcoded paths — only runs on my machine
- No documentation, no version control

This is **extremely common** in economics.

Code Audit: After

Dimension	Before	After
Do-files	51	**8**
Master file	2021	Current
Config system	None	Template
Version control	None	Git
Documentation	None	Full
"Which version?"	Constant	**Zero**

86 old variants preserved in Archive/ — nothing deleted, fully reversible.

```
bd-remote-ed/
code/00_master_repo.do
code/07_Dofiles_academic/
_globals.do, 02_datagen.do,
03_vargen.do (THE one), ...
Archive/ (86 old)
config/config_local_template.do
docs/  justfile  README.md
```

Time: ∼2 Claude Code sessions (a few hours). **By hand:** days, minimum.

The Bug: A Copy-Paste Typo Hiding in Plain Sight

The `$flags` variable list

`$flags` = 17 missing-value indicators used as controls in **every regression**. Defined independently in 5 scripts.

3 of 5 scripts (wrong):

```
... _f_c_engaged _f_c_health
    _c_child_work
```

Missing the `_f_` prefix

→

Correct:

```
... _f_c_engaged _f_c_health
    _f_c_child_work
```

Result: `_c_child_work` appeared *twice* in regressions. The actual missing indicator was left out. Survived 10+ versions across 2+ years. No error, no crash — just quietly wrong. **How AI found it:** cross-referencing every `global flags` definition across all 51 files.

Paper Audit: What Is `/econ-audit`?

Adversarial Review Skill

A reusable skill that reads a paper and acts as a **skeptical referee**.

- Reads full paper (LaTeX, tables, appendix)
- Checks identification, power, measurement
- Produces structured audit report
- Takes ∼60 seconds

```
$ npx skills add thinkingwithagents/skills \
-skill econ-audit
```

What it checks:

1. **Identification:** Is the causal claim valid?
2. **Statistics:** MHT, power, attrition
3. **Measurement:** Are variables measured well?
4. **Methodology:** Design consistency
5. **Presentation:** Framing, length

LIVE DEMO

Running `/econ-audit` on the Paper

```
$ claude
/econ-audit main.tex
```

Reads the full paper + tables + appendix.

Produces a structured audit in ∼60 seconds.

Switching to terminal. . .

AI Audit vs. Real Referees: 7 of 10 in ∼60 Seconds

Referee Criticism	AI?	Notes
High attrition & differential attrition	✔	Flagged as Critical; suggested Lee bounds
Results don't tell a coherent story	◐	Caught ID problem, missed internal contradictions
Theoretical model not useful	✔	Flagged — but recommended *including* it
Short intervention duration (4–8 weeks)	✘	Requires field knowledge
Noisy learning measures (8-item phone test)	✔	Strong match; flagged IRT minimums
Paper too long and hard to follow	✔	Suggested cutting 20–30%
Multiple hypothesis testing issues	✔	Aggressive; added winner's curse framing
Limited external validity / COVID fatigue	◐	Partial; missed publication landscape
Deviations from pre-analysis plan	✔	Caught the spirit, not the specifics
Complex randomization hard to interpret	✔	Strong match

5 full matches, 2 partial, 1 opposite recommendation, 2 misses.
11 referee reports across 8 journals over 4 years — vs. 60 seconds of AI.

What the AI Missed — And Why It Matters

✗ The Misses

- **Short duration:** 4–8 weeks is unusually short for an education RCT. Requires knowing field norms.
- **Internal contradictions:** Data arm increased tutoring *more* than info arm, but only info improved learning. Referees caught this; AI didn't.
- **COVID fatigue:** "We've seen too many of these papers." AI can't know the publication landscape.

The opposite advice:

The AI recommended *including* the conceptual model. Every referee wanted it **cut**.

Sometimes AI gives exactly the wrong recommendation because it lacks field conventions.

Pattern:

AI catches **statistical/econometric** issues reliably.

AI misses **field knowledge**, **within-paper logic checks**, and **publication strategy**.

Takeaway: A Very Good Pre-Submission Check

✔ What It Replaces

- The first 80% of what a careful internal seminar discussant would say
- Catches the systematic stuff: identification, power, MHT, attrition
- In one minute, not one month

What it doesn't replace:

- Field-specific intuition
- "Does this make sense given what we know about Bangladesh?"
- Publication strategy advice
- Colleagues who know your work

For the code audit: it found bugs I didn't find in two years. That alone justified the time.

For the paper audit: run it before you send a paper out. It's free. It catches the embarrassing stuff.

Erkmen: Teaching & Applications

Lecture Builder, Research Brainstorm & Web Scraping

Application 3

Lecture Builder Skill — From Papers to a Lecture

Where We Left Off — The Skill Spec

```
/skill-creator

Create a skill called "lecture-builder" that converts academic documents into lecture materials.  The skill should:

1.  Read documents in their entirety (PDF, DOCX, web pages) -- break reading into parts to avoid excessive token
usage.  Include a Python parsing script.

2.  Extract:  key arguments, empirical findings, methodological details, discussion points, potential exam questions.

3.  Generate two outputs:

a) Structured lecture notes (markdown) with section headers, key takeaways, discussion prompts

b) A polished slide deck (draw on \academic-beamer-deck or \pptx) with clean visuals, one idea per slide

4.  Adapt tone for undergraduate vs. graduate audiences (configurable).

Save SKILL.md + supporting files to .claude/skills/lecture-builder/.
```

Last week we wrote the spec. ***Today we actually build the skill — then run it.***

Building the Skill: `/skill-creator` Walkthrough

Building the Skill: `/skill-creator` Walkthrough

1. Invoke
`/skill-creator`
→
2. Paste
the spec
→
3. Interview
4–6 design Qs
→
4. Skill writes
`SKILL.md` + script
→
5. Saved
`.claude/skills/`

`/skill-creator` **interviews you**. Be ready to answer:
name, audience, inputs, outputs, edge cases, where to save.

Anatomy of `lecture-builder`

`.claude/skills/lecture-builder/`

- `SKILL.md` playbook
- `parse_document.py` PDF/DOCX reader
- `slide_template.tex` UVM Beamer
- `notes_template.md` notes scaffold

What `SKILL.md` enforces

- Chunked reading $\rightarrow$ token control
- Mandatory citations from source PDFs
- Undergrad vs. grad tone toggle
- Dual output: notes + slides
- Reuses UVM color palette

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Skills with Python scripts > pure-markdown skills when tokens matter. The script reads; the LLM synthesizes.

The Input Folder

./inputs/

gov_thinktank_ice_arrests.pdf
policy report

ice_enforcement_report.pdf
government brief

aslim_paper.pdf
academic working paper

Why this mix?

- Heterogeneous sources: **policy** + **government** + **academic**
- Forces the skill to harmonize tone, terminology, and evidence quality
- Mirrors how you'd actually build a lecture

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- Mirrors how you'd actually build a lecture

Topic: The labor and health consequences of ICE enforcement. Connects directly to ongoing research at UVM.

The Execution Prompt

```
/lecture-builder

I have a folder ./inputs/ with:  gov_thinktank_ice_arrests.pdf,
ice_enforcement_report.pdf, aslim_paper.pdf.

Build a 50-minute undergraduate lecture on the labor and health
consequences of ICE enforcement.  Audience:  junior/senior econ majors
who have taken intermediate micro and one applied econometrics class.

Outputs to ./outputs/:
  1) lecture_notes.md -- sectioned notes with 5 key takeaways,
    4 discussion prompts, 3 candidate exam questions.
  2) lecture_slides.tex -- Beamer deck (~20 slides, 16:9).  Use
    a clean academic palette -- default UVM (deep green / gold) or ask
    me.  In-text citations to all three source PDFs.

Be conservative with tokens:  parse each PDF in chunks via
parse_document.py, summarize per-section, then synthesize.
```

Audience spec + output spec + token discipline. Skills work better when the prompt is concrete.

Expected Output (Live Demo Running)

▶ Live demo running in the background — final files appear by the end of this section

lecture_notes.md

 # 1. The setting
Recent expansions of interior immigration enforcement...
 # 2. Identification strategy
287(g) rollout as a staggered DiD...
 # 3. Findings
Aslim et al. document...
 # Discussion prompts
1. Why might the effect on infant health attenuate over time?

 # Exam questions
1. Sketch the parallel-trends test...

synthesizes →

lecture_slides.tex

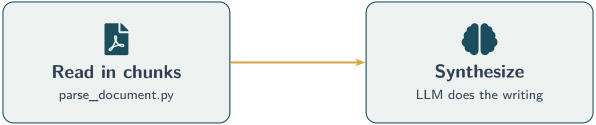
```
% 20 slides, 16:9
% UVM palette
\begin{frame}{Setting}
  ICE arrests timeline
\end{frame}
\begin{frame}{Methods}
  staggered DiD
\end{frame}
\begin{frame}{Results}
  main effect
\end{frame}
```

What Just Happened (Under the Hood)

Read in chunks

parse_document.py

What Just Happened (Under the Hood)



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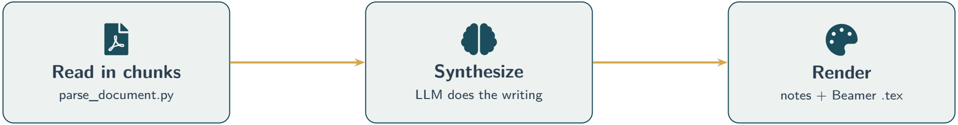
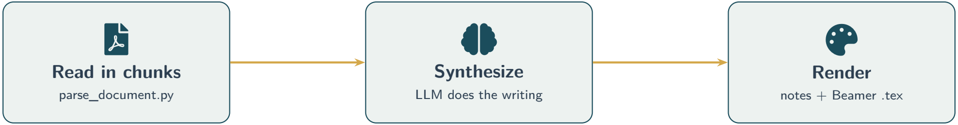


What Just Happened (Under the Hood)



This skill is reusable. Swap the input folder, swap the topic, swap the audience — get a new lecture in minutes. Build once, deploy forever.

Application 4

`/research-brainstorm` — Stress-Test a Research Idea

What the Research Brief Contains

Brief sections

- **The question** — structured: unit, population, treatment, outcome, comparison, setting
- **Closest related work** — 3–6 cites with sharp delta
- **Contribution type** — setting / identification / reframing / measurement / mechanism / policy
- **The strongest objection** — one paragraph, unvarnished
- **Alternative framings** — 2–3 pivots
- **Feasibility** — data, identification, power, timeline, fatal risks
- **Next concrete step**

Saved to:
`research_briefs/YYYY-MM-DD-slug.md`

Markdown so it lives next to your other notes. Re-openable, editable, citeable in your own write-ups.

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The conversation is great.
The artifact is what you keep.

Installing `/research-brainstorm`

Full economist pack

```
npx skills add
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```

Installs the full pack in one go. Today we use: `research-brainstorm`, `find-data`, `academic-beamer-deck`.

One skill at a time

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Mix-and-match per project. Lighter footprint when you only need one.

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Same `npx skills` mechanism works for any GitHub-hosted skill repo. **Build your own pack and share it with your coauthors.**

Live Demo: Hand Me a Question

Audience:

hand me a half-formed research question.

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Audience:

hand me a half-formed research question.

An effect you suspect but can't yet measure	**A dataset** you have access to but haven't used	**A policy** you want to evaluate

Live: invoke `/research-brainstorm`, ride through the phases, walk away with a brief.

Application 5

Web Scraper — From `/find-data` to a Live Visualization

`/find-data` Picks NBER Working Papers

`/find-data` in one line

We hand it our constraints (applied-micro audience, 24+ months, scrape-able, time-series + cross-section). It scans repositories and surfaces a source.

Today's pick: NBER working papers.

Why NBER: every economist cares; clean HTML; rich JEL + affiliation cross-section; ~1,500 papers / year.

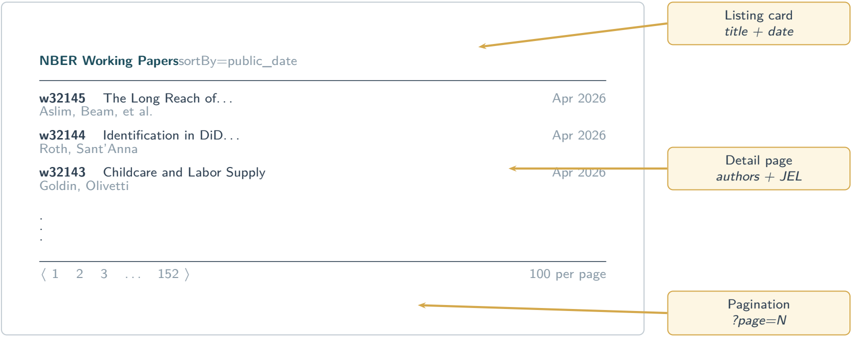
URL:

`nber.org/papers?page=N&perPage=100&sortBy=public_date`

Extract: `nber_number`, `title`, `authors`, `public_date`, `primary_jel_code`, `affiliations`.

Goal: novel data on the screen in under 10 minutes. Skip the buildup — go straight to scraping.

Inspect Before You Prompt



View-source first. The agent writes better scrapers when you've already named the selectors.

The Scraping Prompt

```
Scrape NBER working papers for a teaching demo. Build a self-contained
Python script in ./scrape_nber/ using uv:
1. Fetch the most recent ~1,000 papers from
   nber.org/papers?page=N&perPage=100&sortBy=public_date
   (paginate until 1,000 or cutoff 2024-01-01).
2. Extract: nber_number, title, authors, public_date, primary
   JEL code, affiliations (from detail page if needed).
3. Save to ./scrape_nber/data/nber_papers.csv.
4. Generate three PNGs to ./scrape_nber/figures/:
   - papers_per_month.png   bars, last 24 months, UVM colors
   - top_jel_codes.png   top 15 primary JEL codes
   - top_institutions.png   top 10 affiliations
5. Print 5-line stdout summary.
Politeness: real-browser User-Agent; sleep 0.5s between fetches;
cache HTML to ./scrape_nber/cache/ so re-runs are free.
```

What Claude Does While We Keep Talking

Fetching pages. . .

Parsing. . .

Plotting. . .

```
$ uv run scrape_nber/main.py
[INFO] fetching page 1/10 ...ok (98 papers)
[INFO] fetching page 2/10 ...ok (100 papers)
[INFO] ...
[INFO] parsing JEL codes ...
[INFO] saved data/nber_papers.csv
[INFO] writing figures/...

== Summary ==
n_papers:  1,000
date range:  2024-04 to 2026-04
top JEL: J (Labor)
top affil:  NBER, Harvard, MIT
```

The terminal is a TV. Let it run while you talk through what's coming.

What Claude Does While We Keep Talking

✔ Fetched 10 / 10 pages

Parsing 1,000 records. . .

Plotting. . .

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What Claude Does While We Keep Talking

✔ Fetched 10 / 10 pages

✔ Parsed 1,000 records

Plotting 3 figures. . .

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What Claude Does While We Keep Talking

- ✔ Fetched 10 / 10 pages
- ✔ Parsed 1,000 records
- ✔ Wrote 3 PNGs

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The terminal is a TV. Let it run while you talk through what's coming.

What to Try First

Start Small, Check Each Step

What to Try First

#	Activity	Time	What You Get
1	Create a voice file	15 min	Your tone in every AI draft — reuse forever
2	Data pipeline (describe $\rightarrow$ propose $\rightarrow$ code)	30 min	Orientation on any dataset
3	Install `/econ-audit`, run on a paper	10 min	Pre-submission check for free
4	Try inbox triage with Gmail integration	30 min	Sorted, triaged, draft replies

All four use the same principle: **AI does the mechanical parts, you check each step.** Start with whichever one matches a task you already have.

What We Built Today

1. Data pipeline: curiosity → map → memo in **20 minutes**

What We Built Today

1. Data pipeline: curiosity $\longrightarrow$ map $\longrightarrow$ memo in **20 minutes**
2. Code audit + paper audit: AI matched **7 of 10** real referee criticisms

What We Built Today

1. Data pipeline: curiosity → map → memo in **20 minutes**
2. Code audit + paper audit: AI matched **7 of 10** real referee criticisms
3. Built and ran a real teaching skill — **lecture-builder**

What We Built Today

1. Data pipeline: curiosity $\rightarrow$ map $\rightarrow$ memo in **20 minutes**
2. Code audit + paper audit: AI matched **7 of 10** real referee criticisms
3. Built and ran a real teaching skill — **lecture-builder**
4. Stress-tested a research idea with **evidence-grounded critique**

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Five workflows. Five artifacts on disk. **All reusable.**

Resources

Links

- **Bootcamp website:**
  `thinkingwithagents.github.io`
- **Skills library:**
  `skills.sh`
- **Econ skills:**
  `github.com/thinkingwithagents/skills`
- **Claude Code docs:**
  `docs.anthropic.com/en/docs/claude-code`

Install Commands

Step 1: Install Node.js

`nodejs.org` → click LTS

Step 2: Install Claude Code

```
npm install -g @anthropic-ai/claude-code
```

Step 3: Install econ skills

```
npx skills add thinkingwithagents/skills
```

Step 4: Launch

```
claude
```

Thinking With Agents

Online sessions coming this summer

Workshops, deep-dives, and Q&A on agentic AI for academic research — built by economists, for economists and academics alike.

Stay tuned for dates and registration:

`thinkingwithagents.github.io`